# MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

**END SEMESTER EXAMINATION (2022-23) -- SEMESTER 2**

**COURSE: BASIC ELECTRONICS ENGINEERING (BEC-201)**

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: BEC-201

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

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NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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## SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Draw V-I characteristics of PN junction diode and define knee voltage.

Q1. (b) What is peak inverse voltage (PIV) of center-tapped full wave rectifier?

Q1. (c) Define alpha, beta, and gamma current gain factors in BJT and write their relationship.

Q1. (d) What is pinch-off voltage in JFET?

Q1. (e) State ideal characteristics of an Operational Amplifier.

Q1. (f) What is virtual ground concept in inverting Op-Amp?

Q1. (g) Convert (110101.11)_2 into octal and hexadecimal.

Q1. (h) State De-Morgan laws of Boolean algebra.

Q1. (i) What is Barkhausen criterion for sustained oscillations?

Q1. (j) Differentiate between enhancement and depletion MOSFET.

## SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Explain the working of bridge rectifier with circuit diagram and derive efficiency and ripple factor.

Q2. (b) Explain input and output characteristics of BJT in Common Emitter (CE) configuration with circuit.

Q2. (c) Derive the expression for voltage gain of inverting and non-inverting operational amplifiers.

Q2. (d) Explain construction and transfer characteristics of N-channel JFET with pinch-off derivation.

Q2. (e) Design an Op-Amp based summing amplifier and integrator circuit with output equations.

## SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

**Q3. (a) Explain Zener diode as voltage regulator with line and load regulation circuits. [10 Marks]**

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(b) Draw and explain working of positive and negative clamper circuits with waveforms. [10 Marks]

**Q4. (a) Explain DC load line and Q-point stability in BJT. Explain voltage divider biasing. [10 Marks]**

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(b) Derive relationship between h-parameters and small-signal equivalent model of CE amplifier. [10 Marks]

**Q5. (a) Explain the working of enhancement MOSFET with drain and transfer curves. [10 Marks]**

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(b) Compare BJT, JFET, and MOSFET based on input impedance, temperature stability, and noise. [10 Marks]

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**Q6. (a) Explain working of Op-Amp Schmitt trigger with hysteresis loop and trip points. [10 Marks]**

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(b) Explain Op-Amp differentiator circuit with frequency response and practical modifications. [10 Marks]

**Q7. (a) Simplify Boolean function F(A,B,C,D) = sum m(0,1,2,5,8,9,10,14) + d(7,15) using K-Map. [10 Marks]**

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(b) Explain RC phase shift oscillator with circuit diagram and frequency derivation f = 1/(2 pi RC sqrt(6)). [10 Marks]

## EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

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***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****